

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE PHASE-I & II

TEST # 05

DATE : 29-07-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

ALLEN CAREER INSTITUTE

"SANKALP", CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

"ISHAVASYAM", B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

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MINOR TEST # 05	Topics	DATE : 29 - 07 - 2018
NURTURE COURSE : PHASE – I & II		
PHYSICS	: Kinematics (Motion along straight line and Motion in a plane)	
CHEMISTRY	: Chemical Bonding and Molecular Structure	
BIOLOGY	: Biological Classification, Cell, Cell Division, The living World, Animal Diversity (upto Ctenophora)	

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BEWARE OF NEGATIVE MARKING

- If R is the maximum horizontal range of a particle, then the greatest height attained by it is :-
 (1) R (2) $2R$ (3) $\frac{R}{2}$ (4) $\frac{R}{4}$
- A particle is projected with a velocity v , so that its range on a horizontal plane is twice the greatest height attained. If g is acceleration due to gravity, then its range is :-
 (1) $\frac{4v^2}{5g}$ (2) $\frac{4g}{5v^2}$
 (3) $\frac{4v^3}{5g^2}$ (4) $\frac{4v}{5g^2}$
- During a projectile motion if the maximum height equals to the horizontal range, then the angle of projection with the horizontal is :-
 (1) $\tan^{-1}(1)$ (2) $\tan^{-1}(2)$
 (3) $\tan^{-1}(3)$ (4) $\tan^{-1}(4)$
- A ball is projected upwards. Its acceleration at the highest point is :-
 (1) zero
 (2) directed upwards
 (3) directed downwards
 (4) such as cannot be predicted
- The point from where a ball is projected is taken as the origin of the co-ordinate axes. The x and y components of its displacement are given by $x = 6t$ and $y = 8t - 5t^2$. What is the velocity of projection?
 (1) 6 ms^{-1} (2) 8 ms^{-1}
 (3) 10 ms^{-1} (4) 14 ms^{-1}
- A particle is fired with velocity u making an angle θ with the horizontal. What is the change in velocity when it is at the highest point?
 (1) $u \cos \theta$ (2) u
 (3) $u \sin \theta$ (4) $(u \cos \theta - u)$
- A ball whose kinetic energy is E , is thrown at an angle of 45° with the horizontal, its kinetic energy at the highest point of its flight will be :-
 (1) E (2) $\frac{E}{\sqrt{2}}$ (3) $\frac{E}{2}$ (4) zero
- A body is projected at such an angle that the horizontal range is three times the greatest height. The angle of projection is :-
 (1) $25^\circ 8'$ (2) $33^\circ 7'$
 (3) $42^\circ 8'$ (4) $53^\circ 8'$
- A ball is thrown from a point with a speed v_0 at an angle of projection θ . From the same point and at the same instant a person starts running with a constant speed $v_0/2$ to catch the ball. Will the person be able to catch the ball? If yes, what should be the angle of projection?
 (1) Yes, 60° (2) Yes, 30°
 (3) No (4) Yes, 45°
- A bullet is dropped from the same height when another bullet is fired horizontally. They will hit the ground :-
 (1) simultaneously
 (2) depends on the observer
 (3) one after the other
 (4) none of these
- An aeroplane moving horizontally with a speed of 180 km/hr drops a food packet while flying at a height of 490 m . The horizontal range is :-
 (1) 180 m (2) 980 m
 (3) 500 m (4) 670 m
- A gun fires two bullets at 60° and 30° with the horizontal. The bullets strike at some horizontal distance. The ratio of maximum height for the two bullets is in the ratio :-
 (1) $2 : 1$ (2) $3 : 1$
 (3) $4 : 1$ (4) $1 : 1$

13. Two paper screens A and B are separated by 150 m. A bullet pierces A and then B. The hole in B is 15 cm below the hole in A. If the bullet is travelling horizontally at the time of hitting A, then the velocity of the bullet at A is :-
(Take $g = 10 \text{ ms}^{-2}$)
- (1) $100\sqrt{3} \text{ ms}^{-1}$ (2) $200\sqrt{3} \text{ ms}^{-1}$
(3) $300\sqrt{3} \text{ ms}^{-1}$ (4) $500\sqrt{3} \text{ ms}^{-1}$
14. A projectile can have the same range R for two angles of projection. If t_1 and t_2 be the times of flight in the two cases, then what is the product of two times of flight?
- (1) $t_1 t_2 \propto R^2$ (2) $t_1 t_2 \propto R$
(3) $t_1 t_2 \propto \frac{1}{R}$ (4) $t_1 t_2 \propto \frac{1}{R^2}$
15. Two stones are projected with the same speed but making different angles with the horizontal. Their ranges are equal. If the angle of projection of one is $\pi/3$ and its maximum height is h_1 , then the maximum height of the other will be :-
- (1) $3h_1$ (2) $2h_1$
(3) $h_1/2$ (4) $h_1/3$
16. A projectile is projected with initial velocity $(6\hat{i} + 8\hat{j}) \text{ m/sec}$. If $g = 10 \text{ ms}^{-2}$, then horizontal range is :-
- (1) 4.8 metre (2) 9.6 metre
(3) 19.2 metre (4) 14.0 metre
17. A number of bullets are fired in all possible directions with the same initial velocity u . The maximum area of ground covered by bullets is :-
- (1) $\pi \left(\frac{u^2}{g} \right)^2$ (2) $\pi \left(\frac{u^2}{2g} \right)^2$
(3) $\pi \left(\frac{u}{g} \right)^2$ (4) $\pi \left(\frac{u}{2g} \right)^2$
18. A train of 150 m length is going towards north direction at a speed of 10 ms^{-1} . A parrot flies at a speed of 5 ms^{-1} towards south direction parallel to the railway track. The time taken by the parrot to cross the train is equal to :-
- (1) 12 s (2) 8 s
(3) 15 s (4) 10 s
19. The position x of a particle varies with time (t) as $x = at^2 - bt^3$. The acceleration at time t of the particle will be equal to zero, where t is equal to :-
- (1) $2a/3b$ (2) a/b
(3) $a/3b$ (4) zero
20. A scooter is going towards east at 10 ms^{-1} turns right through an angle of 90° . If the speed of the scooter remains unchanged in taking this turn, the change in the velocity of the scooter is :-
- (1) 20.0 ms^{-1} in south-western direction
(2) zero
(3) 10.0 ms^{-1} in southern direction
(4) 14.14 ms^{-1} in south-western direction
21. A train 100 m long travelling at 40 ms^{-1} overtakes another train 200 m long travelling at 30 ms^{-1} . The time taken by the first train to pass the second train is :-
- (1) 30 s (2) 40 s (3) 50 s (4) 60 s
22. A car starts from rest and moves with constant acceleration. The ratio of the distance covered in the n^{th} second to that covered in n seconds is :-
- (1) $\frac{2}{n^2} - \frac{1}{n}$ (2) $\frac{2}{n^2} + \frac{1}{n}$
(3) $\frac{2}{n} - \frac{1}{n^2}$ (4) $\frac{2}{n} + \frac{1}{n^2}$
23. A train 200 m long crosses a bridge 300 m long. It enters the bridge with a speed of 30 ms^{-1} and leaves it with a speed of 50 ms^{-1} . What is the time taken to cross the bridge ?
- (1) 2.5 s (2) 7.5 s
(3) 12.5 s (4) 15.0 s

24. A train covers the first half of the distance between two stations at a speed of 40 km/h and the other half at 60 km/h. Then its average speed is :-
 (1) 50 km/h
 (2) 48 km/h
 (3) 52 km/h
 (4) 100 km/h
25. A particle's position as a function of time is described as $y(t) = 2t^2 + 3t + 4$. What is the average velocity of the particle from $t = 0$ to $t = 3$ sec?
 (1) 3 m/s
 (2) 6 m/s
 (3) 9 m/s
 (4) 12 m/s
26. An object of mass 10 kg moves at a constant speed of 10 m/s. A constant force, that acts for 4 s on the object, gives it a speed 2 m/s in opposite direction. The force acting on the object is :-
 (1) -3 N
 (2) -30 N
 (3) 3 N
 (4) 30 N
27. The acceleration a (in ms^{-2}) of a body, starting from rest varies with time t (in s) following the equation $a = 3t + 4$. The velocity of the body at time $t = 2$ s will be :-
 (1) 10 ms^{-1}
 (2) 18 ms^{-1}
 (3) 14 ms^{-1}
 (4) 26 ms^{-1}
28. The displacement of a body along x-axis depends on time as $\sqrt{x} = t + 1$. Then the velocity of body :-
 (1) increases with time
 (2) decreases with time
 (3) independent of time
 (4) none of these
29. A food packet dropped from a helicopter rising up with a velocity of 4 ms^{-1} . The velocity of the packet after three seconds will be :-
 (1) 20.4 ms^{-1}
 (2) 25.4 ms^{-1}
 (3) 28.4 ms^{-1}
 (4) 30.4 ms^{-1}
30. A person standing on an escalator takes time t_1 to reach the top of a tower when the escalator is moving. He takes time t_2 to reach the top of the tower when the escalator is standing. How long will he take if he walks up a moving escalator?
 (1) $t_2 - t_1$
 (2) $t_1 + t_2$
 (3) $t_1 t_2 / (t_1 - t_2)$
 (4) $t_1 t_2 / (t_1 + t_2)$
31. A bus starts moving with acceleration 2 ms^{-2} . A cyclist 96 m behind the bus starts simultaneously towards the bus at 20 m/s. After what time will he be able to overtake the bus?
 (1) 4 s
 (2) 8 s
 (3) 12 s
 (4) 16 s
32. Two cars are moving in the same direction with the same speed of 30 km/h. They are separated by 5 km. What is the speed of the car moving in the opposite direction if it meets the two cars at an interval of 4 minutes?
 (1) 15 km/h
 (2) 30 km/h
 (3) 45 km/h
 (4) 60 km/h
33. A thief is running away on a straight road in a jeep moving with speed of 9 ms^{-1} . A policeman chases him on a motor cycle moving at a speed of 10 ms^{-1} . If the instantaneous separation of the jeep from the motor cycle is 100 m, how long will it take for the policeman to catch the thief?
 (1) 1 s
 (2) 19 s
 (3) 90 s
 (4) 100 s
34. A bus is moving with a velocity 10 ms^{-1} on a straight road. A scooterist wishes to overtake the bus in 100 s. If the bus is at a distance of 1 km from the scooterist, with what velocity should the scooterist chase the bus?
 (1) 50 ms^{-1}
 (2) 40 ms^{-1}
 (3) 30 ms^{-1}
 (4) 20 ms^{-1}

35. Two particles start moving from the same point along the same straight line. The first moves with constant velocity v and the second with constant acceleration a . During the time that elapses before the second catches the first, the greatest distance between the particle is :-
- (1) $\frac{v^2}{a}$ (2) $\frac{v^2}{2a}$
 (3) $\frac{2v^2}{a}$ (4) $\frac{v^2}{4a}$
36. Two cars A and B are moving with same speed of 45 km/h along same direction. If a third car C coming from the opposite direction with a speed of 36 km/h meets two cars in an interval of 5 min, the distance of separation of two cars A and B should be :-
- (1) 6.75 km (2) 7.25 km
 (3) 5.55 km (4) 8.35 km
37. A body is moving according to the equation :
 $x = at + bt^2 - ct^2$,
 where x = displacement and a , b and c are constants. The acceleration of body is :-
- (1) $a + 2bt$ (2) $2b + 6ct$
 (3) $2b - 6ct$ (4) $3b - 6ct^2$
38. From the top of the tower of height 400 m, a ball is dropped by a man, simultaneously from the base of the tower, another ball is thrown up with velocity 50 m/s; at what distance will they meet from the base of the tower?
- (1) 100 m (2) 320 m
 (3) 80 m (4) 240 m
39. A car accelerated from initial position and then returned at initial point, then :-
- (1) velocity is zero but speed increases
 (2) speed is zero but velocity increases
 (3) both speed and velocity increase
 (4) both speed and velocity decrease
40. A very large number of balls are thrown vertically upwards in quick succession in such a way that the next ball is thrown when the previous one is at the maximum height. If the maximum height is 5 m, then the number of balls thrown per minute is :- ($g = 10 \text{ ms}^{-2}$)
- (1) 120 (2) 80 (3) 60 (4) 40
41. A stone is released from the top of a tower. If its velocity at half of the height is 10 m/s, then height of the tower is :- ($g = 10 \text{ ms}^{-2}$)
- (1) 8 m (2) 10 m
 (3) 12 m (4) 16 m
42. P is the point of contact of a wheel and the ground. The radius of the wheel is 1 m. The wheel rolls on the ground without slipping. The displacement of point P when the wheel completes half rotation is :-
- (1) 2 m (2) $\sqrt{\pi^2 + 4}$ m
 (3) π m (4) $\sqrt{\pi^2 + 2}$ m
43. An object of mass 3 kg is at rest. If a force $\vec{F} = (6t^2\hat{i} + 4t\hat{j})\text{N}$ is applied on the object, then the velocity of the object at $t = 3$ seconds is :-
- (1) $18\hat{i} + 3\hat{j}$ (2) $18\hat{i} + 6\hat{j}$
 (3) $3\hat{i} + 18\hat{j}$ (4) $18\hat{i} + 4\hat{j}$
44. A body A starts from rest with an acceleration a_1 . After 2 seconds, another body B starts from rest with an acceleration a_2 . If they travel equal distances in the 5th second, after the start of A, then the ratio $a_1 : a_2$ is equal to :-
- (1) 5 : 9 (2) 5 : 7
 (3) 9 : 5 (4) 9 : 7
45. Three different objects of masses m_1 , m_2 and m_3 are allowed to fall from rest and from the same point O along three different frictionless path. The speeds of the three objects on reaching the ground will be in the ratio of :-
- (1) $m_1 : m_2 : m_3$
 (2) $m_1 : 2m_2 : 3m_3$
 (3) 1 : 1 : 1
 (4) $\frac{1}{m_1} : \frac{1}{m_2} : \frac{1}{m_3}$

46. The correct order for bond angles is :-

- (1) $\text{NH}_4^+ > \text{NH}_3 > \text{NH}_2^-$
 (2) $\text{NH}_3 > \text{NH}_4^+ > \text{NH}_2^-$
 (3) $\text{NH}_4^+ > \text{NH}_2^- > \text{NH}_3$
 (4) $\text{NH}_3 > \text{NH}_2^- > \text{NH}_4^+$

47. O_2^{2-} is isoelectronic with :-

- (1) H_2 (2) N_2 (3) F_2 (4) S

48. How many bonded electron pairs are present in IF_7 molecule?

- (1) 6 (2) 7 (3) 5 (4) 8

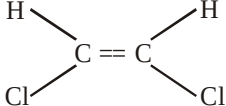
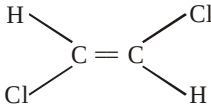
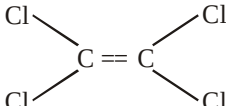
49. PCl_5 exists but NCl_5 does not because :-

- (1) nitrogen has no vacant d-orbitals
 (2) NCl_5 is unstable
 (3) nitrogen atom is much smaller than P
 (4) nitrogen is highly inert

50. Which of the following does not contain co-ordinate bond?

- (1) BH_4^- (2) NH_2^-
 (3) SO_3 (4) H_3O^+

51. Select the isomers given below, which have non-zero dipole moment?

- (1)  (2) 
 (3)  (4) all of these

52. Dipole moment is highest for :-

- (1) Cl_4 (2) CH_4
 (3) CHF_3 (4) CCl_4

53. Bond-length of HCl is :

$1.275 \text{ \AA}$ ($e = 4.8 \times 10^{-10} \text{ esu}$). If $\mu = 1.05 \text{ D}$ then HCl is :-

- (1) 100% ionic (2) 83% covalent
 (3) 50% covalent (4) 40% ionic

54. Which bond angle, θ would result in the maximum dipole moment for the triatomic molecules XY_2 ?

- (1) $\theta = 90^\circ$ (2) $\theta = 120^\circ$
 (3) $\theta = 150^\circ$ (4) $\theta = 180^\circ$

55. Among LiCl , BeCl_2 , BCl_3 and CCl_4 the covalent bond character follows the order :-

- (1) $\text{LiCl} > \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$
 (2) $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$
 (3) $\text{LiCl} > \text{BeCl}_2 > \text{CCl}_4 > \text{BCl}_3$
 (4) $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 > \text{CCl}_4$

56. The correct order of decreasing polarisability of ion is :-

- (1) $\text{Cl}^- > \text{Br}^- > \text{I}^- > \text{F}^-$
 (2) $\text{F}^- > \text{I}^- > \text{Br}^- > \text{Cl}^-$
 (3) $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$
 (4) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$

57. Which ion has a higher polarising power ?

- (1) Mg^{2+} (2) Al^{3+} (3) Ca^{2+} (4) Na^+

58. The bonds present in N_2O_5 are :-

- (1) ionic
 (2) covalent and co-ordinate
 (3) covalent
 (4) ionic and covalent

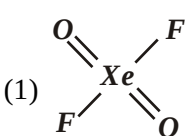
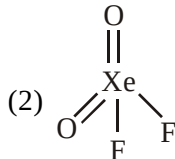
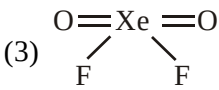
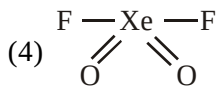
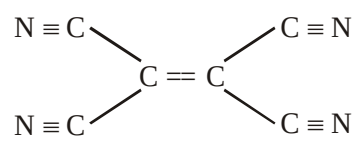
59. The pair having similar geometry is :-

- (1) BF_3 , NF_3 (2) BF_3 , AlF_3
 (3) BeF_2 , H_2O (4) BCl_3 , PCl_3

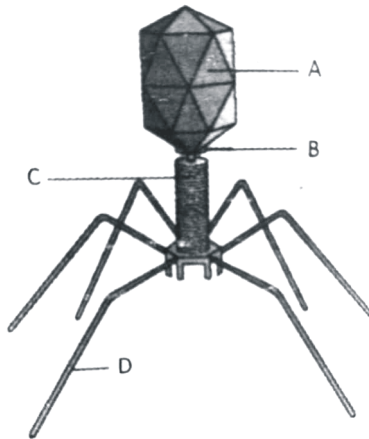
60. The shape of ClO_3^- according to VSEPR model is :-

- (1) planar triangular (2) pyramidal
 (3) tetrahedral (4) square planar

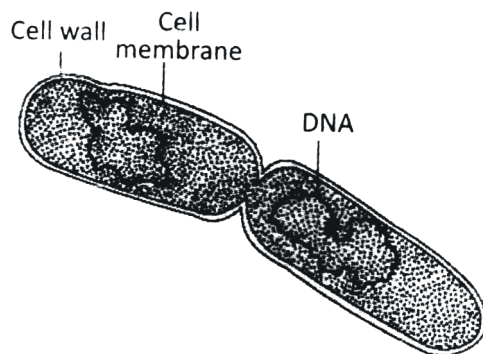
61. The correct order for bond angles is :-
 (1) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$
 (2) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_2$
 (3) $\text{NO}_2 > \text{NO}_2^- > \text{NO}_2^+$
 (4) $\text{NO}_2^- > \text{NO}_2 > \text{NO}_2^+$
62. The shapes of PCl_4^+ , PCl_4^- and AsCl_5 are respectively :-
 (1) square planar, tetrahedral, see-saw
 (2) tetrahedral, see-saw, trigonal bipyramidal
 (3) tetrahedral, square planar and pentagonal bipyramidal
 (4) trigonal bipyramidal, tetrahedral and square pyramidal
63. Which of the following two are isostructural?
 (1) XeF_2 , IF_2^- (2) NH_3 , BF_3
 (3) CO_3^{2-} , SO_3^{2-} (4) PCl_5 , ICl_5
64. H-B-H bond angle in BH_4^- is :-
 (1) 180° (2) 120° (3) 109° (4) 90°
65. The correct hybridisation state of sulphur atom in SF_2 , SF_4 and SF_6 molecules is respectively :-
 (1) sp^3d , sp^3 , sp^3d^2 (2) sp^3 , sp^3d , sp^3d^2
 (3) sp^3d^2 , sp^3 , sp^3d (4) sp^3d^2 , sp^3d , sp^3
66. Which molecule is T-shaped?
 (1) BeF_2 (2) BCl_3 (3) NH_3 (4) ClF_3
67. The hybridisation of P in phosphate ion (PO_4^{3-}) is the same as in :-
 (1) I in ICl_4^- (2) S in SO_3
 (3) N in NO_3^- (4) S in SO_3^{2-}
68. How many unpaired electrons are present in N_2^+ ?
 (1) 1 (2) 2 (3) 3 (4) 4
69. The diamagnetic molecules are :-
 (1) B_2 , C_2 , N_2 (2) O_2 , N_2 , F_2
 (3) C_2 , N_2 , F_2 (4) B_2 , O_2^{2-} , N_2
70. The correct bond order for CO and CO^+ are respectively :-
 (1) 3, $5/2$ (2) 3, 2 (3) 3, $7/2$ (4) $5/2$, 3
71. Bond order of NO is :-
 (1) 1.0 (2) 1.5 (3) 2.0 (4) 2.5
72. H-bonding is not present in :-
 (1) glycerine (2) water
 (3) H_2S (4) HF
73. Which one of the following molecules show banana bonding?
 (1) BeH_2 (2) BH_3
 (3) BF_3 (4) Both (1) and (2)
74. The correct order of increasing bond angles is:-
 (1) $\text{PF}_3 < \text{PCl}_3 < \text{PBr}_3 < \text{PI}_3$
 (2) $\text{PF}_3 < \text{PBr}_3 < \text{PCl}_3 < \text{PI}_3$
 (3) $\text{PI}_3 < \text{PBr}_3 < \text{PCl}_3 < \text{PF}_3$
 (4) $\text{PF}_3 > \text{PCl}_3 < \text{PBr}_3 < \text{PI}_3$
75. In the series ethane, ethylene and acetylene, the C-H bond energy is :-
 (1) the same in all the three compounds
 (2) greatest in ethane
 (3) greatest in ethylene
 (4) greatest in acetylene
76. Which of the following possess lowest bond energy?
 (1) C-C (2) N-N
 (3) H-H (4) O-O
77. The angle between two covalent bonds is maximum in :-
 (1) CH_4 (2) H_2O (3) CO_2 (4) SO_3
78. Which of the following has zero dipole moment?
 (1) CH_2Cl_2 (2) CH_4
 (3) NH_3 (4) PH_3

79. Which molecule is planar?
 (1) CH_4 (2) C_2H_6
 (3) NH_3 (4) C_6H_6
80. When the hybridisation state of carbon atom changes from sp^3 to sp^2 to sp , the angle between the hybridised orbitals :-
 (1) decreases gradually
 (2) increases gradually
 (3) decreases considerably
 (4) all of these
81. The shape of IF_7 molecule is :-
 (1) Octahedral
 (2) pentagonal bipyramidal
 (3) Trigonal bipyramidal
 (4) Tetrahedral
82. The bond angle is minimum in :-
 (1) H_2Te (2) H_2Se
 (3) H_2O (4) H_2S
83. In an octahedral structure, the pair of d-orbitals involved in sp^3d^2 -hybridisation is :-
 (1) $\text{d}_{x^2}, \text{d}_{xz}$ (2) $\text{d}_{xy}, \text{d}_{yz}$
 (3) $\text{d}_{x^2-y^2}, \text{d}_{z^2}$ (4) $\text{d}_{xz}, \text{d}_{x^2-y^2}$
84. In a regular octahedral molecule, MX_6 , the number of X-M-X bonds at 180° is :-
 (1) 12 (2) 4
 (3) 3 (4) 6
85. The shape of SO_4^{2-} ion is :-
 (1) square planar
 (2) tetrahedral
 (3) trigonal bipyramidal
 (4) hexagonal
86. Shape of XeO_2F_2 is correctly represented as :-
 (1)  (2) 
 (3)  (4) 
87. HCl is a gas but HF is a low boiling liquid. This is because :-
 (1) H - F bond is strong
 (2) H - F bond is weak
 (3) molecules of HF aggregate because of hydrogen bonding
 (4) HF is a weak acid
88. The number of ionic, covalent and coordinate bonds in NH_4Cl are respectively :-
 (1) 1, 3 and 1
 (2) 1, 3 and 2
 (3) 1, 2 and 3
 (4) 1, 1 and 3
89. The correct order of increasing C - O bond length of CO , CO_3^{2-} , CO_2 is :-
 (1) $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$
 (2) $\text{CO}_2 < \text{CO}_3^{2-} < \text{CO}$
 (3) $\text{CO} < \text{CO}_3^{2-} < \text{CO}_2$
 (4) $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
90. How many σ and π bonds are there in the molecule of tetracyanoethylene?

 (1) Nine σ and nine π
 (2) Five σ and nine π
 (3) Nine σ and seven π
 (4) Five σ and eight π

91. In a bacteriophage A-D are labelled. Select the correct information :-



- (1) A-Head, B-Tail fibres
(2) A-Head, D-Sheath
(3) B-Collar, D-Tail fibres
(4) C-Sheath, A-Collar
92. Malarial parasite is member of :-
(1) Eubacteria (2) Sporozoa
(3) Euglenoids (4) Slime moulds
93. Reproduction in bacteria is shown in following diagram. Select the correct match :-



Mode of Reproduction

- (1) Budding
(2) Binary fission
(3) Spore formation
(4) Budding

Conditions

- Favourable
Favourable
Unfavourable
Unfavourable

94. Dispersal of spores of slime moulds occur through :-
(1) Air (2) Water
(3) Insects (4) Crustaceans
95. Identify the incorrect information about given fungi :-

- (1) Puccinia – It causes wheat rust
(2) Neurospora – Extensively used in genetics
(3) Claviceps – Source of LSD
(4) Ustilago – It is known as rust fungi

96. Deuteromycetes is also known as fungi imperfecti because :-

- (1) Members are not well differentiated
(2) Mycellium is aseptate and coenocytic
(3) Members do not produce zoospores
(4) It lacks sexual reproduction

97. Presence of some bacteria is expected in hot springs. These are :-

- (1) Archaeobacteria
(2) Phototrophic autotrophs
(3) Chemosynthetic autotrophs
(4) Heterotrophic eubacteria

98. Identify the correct statement :-

- (1) All plants without any exception are obligate autotrophs
(2) All fungi without any exception are obligate saprotrophs
(3) All animals without any exception are obligate heterotrophs
(4) All monerans without any exception are obligate anaerobes

99. Identify the incorrect match :-

Class of fungi	Sexual/Asexual spores	Example
(1) Phycomycetes	Oospores	Rhizopus
(2) Ascomycetes	Ascospores	Neurospora
(3) Basidiomycetes	Conidia	Agaricus
(4) Deuteromycetes	Conidia	Alternaria

100. Who has given the term virus ?

- (1) Beijerinck (2) Pasteur
(3) Stanley (4) Ivanowsky

101. How many of the following spores are not meant for asexual reproduction in fungi?

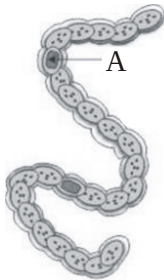
[Conidia, Zoospores, Basidiospores, Ascospores, Oospores, Alphanospores]

- (1) Two (2) Three (3) Four (4) Five

102. Viroid can be defined as :-

- (1) ds DNA with protein coat
- (2) free RNA
- (3) ssRNA with capsid
- (4) free DNA

103. Diagram of Nostoc is given. Select the correct statement :-



- (1) A is mucilaginous sheath and is also found in Rivularia
- (2) A is mucilaginous sheath and is also found in Anabaena
- (3) A is heterocyst and is also found in methanogens
- (4) A is heterocyst and is also found in Anabaena

104. Identify the incorrect statement :-

- (1) Conidia in ascomycetes are exogenously produced
- (2) Basidiospores in mushrooms are exogenously produced
- (3) Zoospores in Mucor are endogenously produced
- (4) Ascospores in sac fungi are exogenously produced

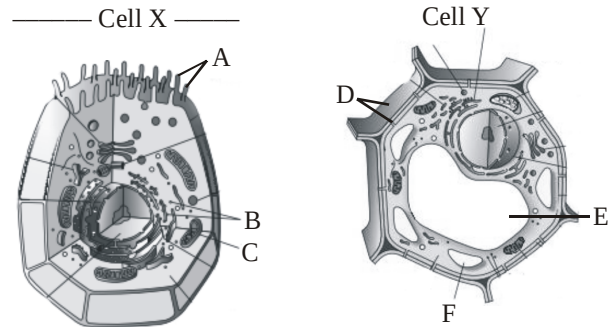
105. Select the option with edible fungi :-

- (1) Buffles only
- (2) Truffles only
- (3) Both buffles and morels
- (4) Both morels and truffles

106. Which of the following is not correctly matched with his contribution?

- (1) Theodore Schwann : Cell theory
- (2) Singer and Nicolson : Lipoprotein bilayer model
- (3) Robert Brown : Discovery of nucleus
- (4) Kolliker : Discovery of mitochondria

107. Select the option with correct identification of cell type with one of the features given :-



- (1) Cell X is an animal cell and A and B are found in animal cell only
- (2) Cell X is a plant cell and A and C are found in plant cell only
- (3) Cell Y is an animal cell and D and E are found in animal cell only
- (4) Cell Y is a plant cell and D is found in plant cell only

108. Select the incorrect statement :-

- (1) Mannans and galactans are found in cell membrane of algae
- (2) Electron microscope was discovered by Knoll and Ruska
- (3) Inclusion bodies are not bound by cell membrane
- (4) Vibrio are comma shaped bacteria

109. Out of the following groups of organisms, how many are prokaryotic?

A-Fungi, B-Protista, C-Algae, D-Mycoplasma, E-Blue green algae

- (1) 1 (2) 2 (3) 3 (4) 4

110. Incorrect about Na⁺-K⁺ ATPase pump is :-

- (1) Utilize ATP
- (2) Transports Na⁺ into the cell
- (3) Transport K⁺ into the cell
- (4) Present in sarcolemma

111. Select the incorrect statement about the cell wall of plant cell :-

- (1) It is a non-living structure
- (2) It is differentially permeable
- (3) It can be distinguished into primary cell wall, secondary cell wall and middle lamella
- (4) It consists mainly of cellulose and hemicellulose

112. Select the incorrect statement about integral proteins of cell membrane?

- (1) These are tightly bound to lipid bilayer
- (2) These are very difficult to remove from the membrane
- (3) Cytochrome P₄₅₀ is an example of integral protein
- (4) These cannot be removed by use of detergents and organic solvents

113. Which cell organelle is not matched with its main function?

- (1) Smooth endoplasmic reticulum : Steroid synthesis
- (2) Ribosomes : Protein synthesis
- (3) Golgi bodies : Degradation of glycolipids
- (4) Centrioles : Cell division

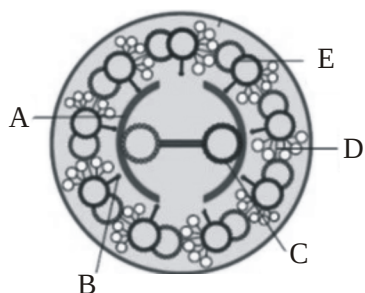
114. pH inside lysosome is :-

- (1) Acidic, 5 (2) Acidic, 6.8
- (3) Alkaline, 7.4 (4) Alkaline, 8.8

115. If A = Nucleus, B=Plastid, C = Mitochondria and D = Vacuole, then arrange the given structures on the basis of their size :-

- (1) A > B > C > D (2) D > C > B > A
- (3) D > A > B > C (4) D > B > A > C

116. Section of a centriole is given. Select the incorrect option :-



- (1) B is radial spoke and it connects the central sheath with peripheral microtubule
- (2) D is inter-doublet bridge and it connects the two peripheral microtubules
- (3) C is central sheath and it covers the central microtubules
- (4) A is central sheath and E is peripheral doublet

117. Oxysomes in mitochondria can be observed in :-

- (1) Outer mitochondrial membrane
- (2) Inner mitochondrial membrane
- (3) Peri-mitochondrial space
- (4) Matrix

118. Which of the following statement is correct about mesosomes.

- (1) They can be seen in both gram positive and gram negative bacteria
- (2) They may be involved in cell wall formation and cell division
- (3) They play a role in chromosome replication and distribution to daughter cells.
- (4) All of these

119. Typical human cell divides in every :-

- (1) 12 Hrs (2) 24 Hrs
- (3) 1 Week (4) 1 Month

120. Which pair of body cells lack cell division?

- (1) Skin epithelial cells and nephrons
- (2) Nephrons and endothelial cells
- (3) Gut lining cell and neurons
- (4) Neurons and heart cells

121. Which of the following events is not related to mitotic prophase?

- (1) Spread of chromosome in the cytoplasm
- (2) Disappearance of nuclear membrane
- (3) Assembly of mitotic spindle
- (4) Condensation of chromatin material

122. If Diplotene - P, Diakinesis - Q, Leptotene -R, Zygotene - S and Pachtene - T, then arrange them in the sequence of occurrence.

- (1) P → R → S → Q → T
- (2) R → P → T → S → Q
- (3) R → Q → S → P → T
- (4) R → S → T → P → Q

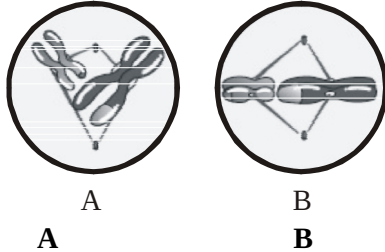
123. Genetic variability in sexually reproducing organisms is due to _____.

- (1) Amitosis (2) Multiple fission
- (3) Mitosis (4) Meiosis

124. In meiosis during prophase-I, recombination nodules are observed during :-

- (1) Zygotene (2) Diakinesis
(3) Diplotene (4) Pachytene

125. Identify the option with correct identification of given stages of the cell division (A and B).



- (1) Prophase Metaphase
(2) Prophase-I Metaphase-II
(3) Prophase-II Metaphase-I
(4) Prophase-II Metaphase-II

126. Select the correct statement :-

- (1) Metaphasic chromosome in mitosis is having 2 chromatids
(2) Animal cells divide by cell plate method
(3) Reduction of chromosome number occur in meiosis-II
(4) Crossing over is mediated by phosphorylase enzyme.

127. Crossing over is the exchange of genetic material between the :-

- (1) Non-sister chromatids of non-homologous chromosomes
(2) Sister chromatids of non-homologous chromosomes
(3) Non-Sister chromatids of homologous chromosomes
(4) Sister chromatids of homologous chromosomes

128. Identify the correct sequence of stages in a cell cycle :-

- (1) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
(2) $S \rightarrow G_1 \rightarrow G_2 \rightarrow M$
(3) $G_1 \rightarrow G_2 \rightarrow M \rightarrow S$
(4) $M \rightarrow G_1 \rightarrow G_2 \rightarrow S$

129. Most of the bacteria obtain nutrition from :-

- (1) Sunlight
(2) Inorganic substrates
(3) Dead or decaying matter
(4) Plants

130. Which of the following is not applicable to zoological park?

- (1) In vivo mode of conservation
(2) Wild animals are kept under human care
(3) Wild animals are kept in separate enclosures
(4) All are true

131. Which of the following statement is false?

- (1) No non-living object exhibits metabolism
(2) No non-living object shows reproduction
(3) No non-living object shows growth
(4) none is false, all are true statements

132. In which organism reproduction can be considered as synonymous with growth?

- (1) Amoeba (2) Planaria
(3) Star fish (4) Two of these

133. Taxonomy + Relationship between organisms = _____

- (1) Systematics (2) Nomenclature
(3) Cytotaxonomy (4) Classification

134. In a biological name - *Mangifera indica* Linn., Linn indicates :-

- (1) Subspecies (2) Breed
(3) Variety (4) Author

135. Key is used as a taxonomic aid for identification of plants and animals. What is correct for the key?

- (1) Couplet is the pair of contrasting characters
(2) Keys can be used for different taxons
(3) Each statement in the key is termed as lead
(4) All of these

136. Segmentation is present in :-

- (1) Annelida (2) Arthropoda
(3) Chordata (4) All of the above

137. Incomplete digestive system is found in :-

- (1) Arthropoda (2) Annelida
(3) Platyhelminthes (4) Aschelminthes

138. Which of the following is acoelomate ?

- (1) Echinodermata (2) Chordata
(3) Platyhelminthes (4) Both (1) and (2)

139. Triploblastic animals primarily shows :-

- (1) Radial symmetry
(2) Bilateral symmetry
(3) Sexual dimorphism
(4) True coelom

140. Triploblastic organisation and bilateral symmetry starts from which phylum during evolution :-

- (1) Porifera (2) Coelentrata
(3) Aschelminthes (4) Platyhelminthes

141. Which one of the following statements is incorrect?

- (1) Mesoglea is present in between ectoderm and endoderm in Cnidaria
(2) Radial symmetry is found in cnidaria
(3) Platyhelminthes are pseudocoelomate animals
(4) Earthworm is a triploblastic animal

142. Bilaterally symmetrical deuterostome belongs to:-

- (1) Echinodermata (2) Arthropoda
(3) Mollusca (4) Annelida

143. During embryonic development if anus is formed firstly and then mouth is formed, the condition is called :-

- (1) Protostomia (2) Deuterostomia
(3) Both (1) and (2) (4) Astomata

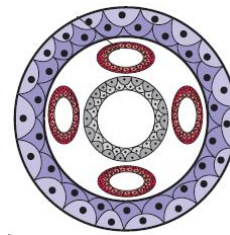
144. Protozoans are belived to be primitive relatives of animals because :-

- (1) All protozoan are autotrophic
(2) Protozoan are primitive multicellular animals
(3) All protozoans are heterotroph and live as predator or parasite
(4) Pellicle makes their body flexible

145. Mark the correct statement :-

- (1) Notochord is ventral whereas nerve cord is dorsal in chordates
(2) Nerve cord is ventral in chordates
(3) Notochord is dorsal in chordates
(4) Notochord is ectodermally derived in chordates

146. Such a character is seen in :-



- (1) Platyhelminthes
(2) Aschelminthes
(3) Annelida
(4) Arthropoda

147. True statement for parazoans :-

- (1) Multicellularity
(2) Prokaryotes
(3) Have tissue level of organisation
(4) Triploblastic

148. Choose correct statement :-

- (1) Asymmetry is seen in all sponges
(2) Echinoderms are Enterocoelomates
(3) Echinoderms have prostomic evolutionary trend
(4) Arthropods have closed circulatory system without any exception

149. In sponges spongocoel is lined by :-

- (1) Porocytes (2) Choanocytes
(3) Pinacocytes (4) Archaeocytes

150. Only sexual reproduction is present in which of the following animal :-

- (1) Obelia (2) Sycon
(3) Pleurobrachia (4) Paramoecium

151. Metagenesis is present in :-

- (1) *Obelia* and *Hydra*
- (2) *Adamsia* and *Physalia*
- (3) *Aurelia* and *Obelia*
- (4) *Physalia* and *Obelia*

152. African sleeping sickness is caused by :-

- (1) *Leishmania donovoni*
- (2) *Giardia intestinalis*
- (3) *Trypanosoma gambiense*
- (4) *Trypanosoma cruzi*

153. Contractile vacuoles are present in :-

- (1) Fresh water protozoan
- (2) Marine water protozoan
- (3) Parasitic protozoan
- (4) Both (1) and (2)

154. Nervous tissue of coelenterate is :-

- (1) Diffuse type with non polar neuron
- (2) Consist of nerve ring and nerve cord
- (3) Diffuse type with bipolar neuron
- (4) Ladder like

155. In hydra hypostome functions as :-

- (1) Mouth
- (2) Anus
- (3) Both
- (4) None

156. Given below is a list of some groups of animal kingdom :

Arthropods, Platyhelminthes, Annelids, Echinoderms, Vertebrates, Cephalopod Mollusc, Nematodes, Porifera and Ctenophora

How many among them have closed circulatory system :-

- (1) 5
- (2) 3
- (3) 4
- (4) All

157. Which statement is not incorrect for sponges ?

- (1) Sponges are mostly sessile, some are motile
- (2) All sponge are found in marine water
- (3) Sponges are mostly radial symmetrical
- (4) Sponges have numerous ostia and single osculum

158. Digestion in coelenterate is :-

- (1) Extracellular only
- (2) Intracellular only
- (3) First extracellular then intracellular
- (4) First intracellular then extracellular

159. Comb plates in sea walnut helps in :-

- (1) Respiration
- (2) Locomotion
- (3) Digestion
- (4) Reproduction

160. The most notorious sporozoan is A which cause B a disease which has staggering effect on human population :-

- (1) A-*Nosema*, B-Pabrine
- (2) A-*Monocystis*, B-Sterility
- (3) A-*Plasmodium*, B-Malaria
- (4) A-*Leishmania* (Leishmania), B-Kala Azar

161. Radial symmetrical, eumetazoan having tissue grade of organisation present in :-

- (1) *Hydra*
- (2) *Ctenoplana*
- (3) Both (1) and (2)
- (4) *Star fish*

162. The protozoan having shell of silica or CaCO_3 belong from :-

- (1) Sporozoan
- (2) Amoeboid protozoan
- (3) Flagellated protozoan
- (4) Ciliated protozoan

163. Fertilisation in porifera :-

- (1) Self and internal
- (2) Cross and external
- (3) Cross and internal
- (4) Self and external

164. Water canal system in sponges helps in :-

- (1) Nutrition
- (2) Respiration
- (3) Excretion
- (4) All

165. Given below is a table of some phyla of animal kingdom and examples, match the correct one :-

	Animals	Common name	Phylum
(1)	<i>Adamsia</i>	Sea anemone	Coelenterata
(2)	<i>Pleurobrachia</i>	Jelly fish	Porifera
(3)	<i>Leucosolenia</i>	Bath sponge	Ctenophora
(4)	<i>Meandrina</i>	Sea pen	Porifera

166. Which cells are present in ctenophora in the place of cnidoblasts of coelenterata ?

- (1) Cnidocyte
- (2) Lasso cell
- (3) Colloblast
- (4) Both (2) and (3)

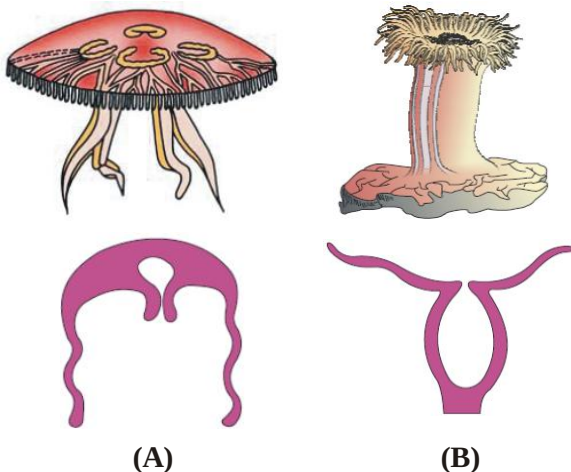
167. Non-chordates are :-

- (1) Porifera to chordata
- (2) Porifera to hemichordata
- (3) Porifera to mollusca
- (4) Porifera to arthropoda

168. If the body of an animal externally and internally divided into segments with a serial repetition of at least some organ, this is called :-

- (1) Pseudometameric segmentation
- (2) Metameric segmentation
- (3) Pseudometamerism
- (4) False segmentation

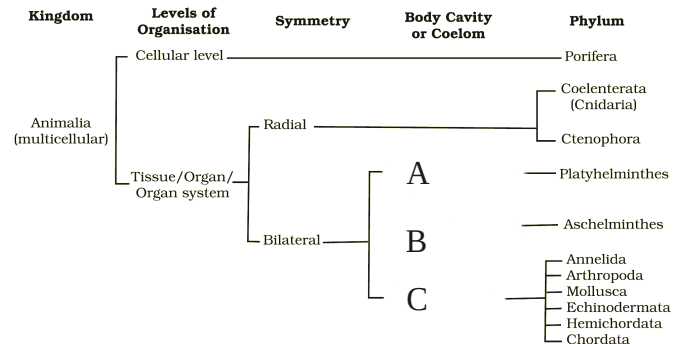
169. Given below is a figure of an animal indicating outline of their body form :-



Choose the correct answer :-

- (1) A-Adamsia (Polyp), B-Aurelia (Medusa)
- (2) A-Aurelia (Medusa), B-Adamsia (Polyp)
- (3) A-Aurelia (Polyp), B-Adamsia (Medusa)
- (4) A-Adamsia (Medusa), B-Aurelia (Medusa)

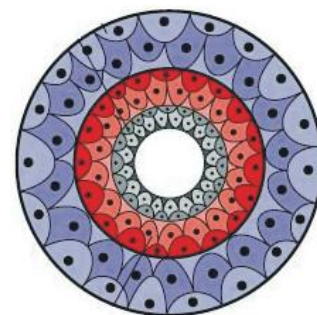
170. A broad classification of kingdom animalia is given below labelled A, B and C :



Find out the correct option :-

- (1) A-Pseudocoelomates, B-Coelomates, C-Acoelomates
- (2) A-Acoelomates, B- Pseudocoelomates, C-Coelomates
- (3) A-Coelomates, B-Acoelomates, C-Pseudocoelomates
- (4) A-Acoelomates, B-Coelomates, C-Pseudocoelomates

171. The cross section of animal body is given below :-



Which of the following group will satisfy the above cross section :-

- (1) Porifera
- (2) Coelenterata
- (3) Ctenophora
- (4) Annelida

172. The phylum which exhibits radial or bilateral symmetry depending on the stage is :-

- (1) Mollusca
- (2) Hemichordata
- (3) Echinodermata
- (4) Ctenophora

173. The body of sponge is supported by a skeleton made up of :-

- (1) Spicule
- (2) Spongin fibres
- (3) Both (1) and (2)
- (4) Coral

174. Match the following column A and B :-

	Column-A		Column-B
(A)	Sea pen	(i)	<i>Ctenoplana</i>
(B)	Sea walnut	(ii)	<i>Adamsia</i>
(C)	Sea fan	(iii)	<i>Pennetula</i>
(D)	Sea anemone	(iv)	<i>Gorgonia</i>

- (1) A-iii, B-i, C-ii, D-iv
- (2) A-iv, B-iii, C-ii, D-i
- (3) A-iv, B-ii, C-iii, D-i
- (4) A-iii, B-i, C-iv, D-ii

175. Which one of the following option is correct regarding to phylum and its character ?

	Figure	Phylum	Character
(1)		Coelenterata	1. Cnidoblast present 2. Medusa form
(2)		Ctenophora	1. Comb plates present 2. Numerous osculum
(3)		Porifera	1. Spicules present 2. Bioluminescence
(4)		Protozoa	1. Flagellated 2. Shell present

176. The undifferentiated cells of sponges that produce the sex cells as well :-

- (1) Thesocytes
- (2) Myocytes
- (3) Archaeocytes
- (4) Chromocytes

177. Body does not show any segmentation in :-

- (1) Frog
- (2) Cockroach
- (3) Earthworm
- (4) Star fish

178. Bioluminescence is well marked character of :-

- (1) Coelenterata
- (2) Porifera
- (3) Ctenophora
- (4) Platyhelminthes

179. What is correct about metagenesis :-

- (1) It is alternation of generation in coelenterates.
- (2) Medusae form the polyps sexually
- (3) Both polyp and medusae form exist.
- (4) All of these

180. Which of the following statement is incorrect about members of phylum - ctenophora

- (1) They are also known as sea walnuts or comb jellies
- (2) Eight ciliated comb plates help them in locomotion.
- (3) Fertilization is internal and indirect development
- (4) monoecious conditions is found

Space for Rough Work

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